Building Reliable Conversational Agents with Genie

A shared methodology for brand performance management across Kerendia, Lynkuet, and Nubeqa -- enabling commercial analytics teams to design, curate, and operate production-grade Genie Agents (Spaces).

4 Hours | Beginner to Intermediate | SQL + UI Walkthrough

INSTRUCTOR

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THE BUSINESS IMPERATIVE

Scaling Pharma Commercial Analytics

Bayer's US Pharma Commercial organization manages a growing portfolio of brands, each with distinct strategies, engagement models, and data requirements.

Prescriber Performance

NBRx, NPS, TRx, prescription funnels, channel utilization, and shifts in prescriber behavior

Field Force Productivity

Call-plan execution, detail frequency, sample drops, and interaction quality

Omnichannel Engagement

P2P programs, digital touchpoints, rep-driven interactions, and hospital-system engagement

Brand Performance

Rx funnels (Submitted -> Paid -> Paid-Insured), competitive dynamics, and territory trends

Workshop Roadmap

Module	Topic	Activity
1	The Challenge -- Scaling Pharma Commercial Analytics	Problem Framing
2	The Solution -- How Genie Spaces Address the Challenge	Concept
3	Lab Setup -- Pharma Commercial Dataset Design	Lab 1: Build the Dataset
4	Understanding Your Data -- Profiling & Quality	Lab 2: Data Profiling
5	Designing a Genie Space -- Planning & Approach	Design Exercise
6	Creating a Genie Space -- Step-by-Step	UI Walkthrough
7	Knowledge Store Curation -- The Full Toolkit	Labs 3-7
8	Benchmarking & Quality Assurance	Lab 8: Benchmark Queries
9	Sharing & Governance	Concept + Walkthrough
10	Monitoring & Continuous Improvement	Lab 9: Usage Analytics

PROGRESSIVE JOURNEY

Beginner to Expert Learning Path

01 Beginner

Understand what Genie is, why governance matters, and how natural language becomes SQL-backed answers for pharma commercial questions.

02 Intermediate

Design a reliable Genie Agent (Space), choose trusted tables, and curate the Knowledge Store with metadata, synonyms, and pharma-specific examples.

03 Advanced

Benchmark answer quality against expected Rx and field activity metrics, share safely, and monitor real-world usage for continuous improvement.

Prerequisites:

Databricks workspace with Unity Catalog, SQL warehouse (Serverless recommended), familiarity with SQL, pharma commercial data concepts.

MODULE 1

Genie Fundamentals

A Genie Agent (Space) is a governed conversational analytics experience in Databricks AI/BI that lets business users ask natural language questions and receive SQL-backed answers.

What Commercial Teams Ask

- "How many NBRx did Kerendia generate this quarter?"
- "Which reps have the highest call-plan execution rate?"
- "What is the NPS trend for Nubeqa by territory?"
- "Show me P2P program ROI by format type"

How Genie Answers

- Interprets the question using curated Knowledge Store context
- Maps the prompt to governed Unity Catalog tables and columns
- Generates and executes SQL against authorized data
- Returns a trustworthy, explainable answer

MODULE 2

Where Genie Fits in AI/BI

SQL Warehouses

Execute the generated SQL queries

Unity Catalog

Govern data, permissions, lineage & metadata

Dashboards

Curated visual consumption layer

Genie Agents (Spaces)

Conversational self-service analytics

MODULE 2

Why Reliability Matters

A Genie Agent (Space) is only useful when it produces answers that meet four critical standards.

Accurate

SQL logic reflects the business definition precisely -- NBRx excludes Switch and Refill

Consistent

Similar prompts resolve to the same meaning every time across all three brands

Governed

Users only see data they are authorized to access via Unity Catalog

Explainable

Output traces back to understandable tables and auditable SQL

MODULE 2

End-to-End Execution Pattern

From natural language question to governed answer in six steps.

1. User asks a business question in natural language
2. Genie identifies the relevant semantic context from the Knowledge Store
3. Genie maps the prompt to tables, columns, business rules, and examples
4. Genie generates SQL against Unity Catalog-governed data
5. The SQL executes on a SQL warehouse
6. The answer is returned with governance automatically enforced

MODULE 3

Pharma Commercial Dataset Design

A compact pharma commercial dataset with six related tables that mirror the real data environment.

Table	Real-World Equivalent	Grain
pharma_lab_hcps	HCP master from MDM / IQVIA OneKey	One row per HCP
pharma_lab_territories	Territory alignment from Veeva CRM	One row per territory
pharma_lab_prescriptions	Rx claims from prescription / specialty-pharmacy feeds	One row per Rx event
pharma_lab_field_activities	Field activity logs from Veeva CRM	One row per interaction
pharma_lab_speaker_programs	P2P programs from PeerDirect / speaker bureau	One row per P2P event
pharma_lab_digital_touchpoints	Vendor feeds from SFMC / Medscape / PulsePoint	One row per digital engagement

40	10	200	120	30	150
HCPS	TERRITORIES	PRESCRIPTIONS	FIELD ACTIVITIES	SPEAKER PGMS	DIGITAL TOUCHES

MODULE 3

Streamlined Lab Flow

1. Create and validate the pharma commercial dataset
2. Profile the data and document business meaning
3. Design the Genie Agent (Space) before building it
4. Create the Genie Agent (Space) over governed tables
5. Curate the Knowledge Store in progressive layers
6. Benchmark response quality with expected answers
7. Share to users through Databricks One
8. Monitor usage and continuously improve

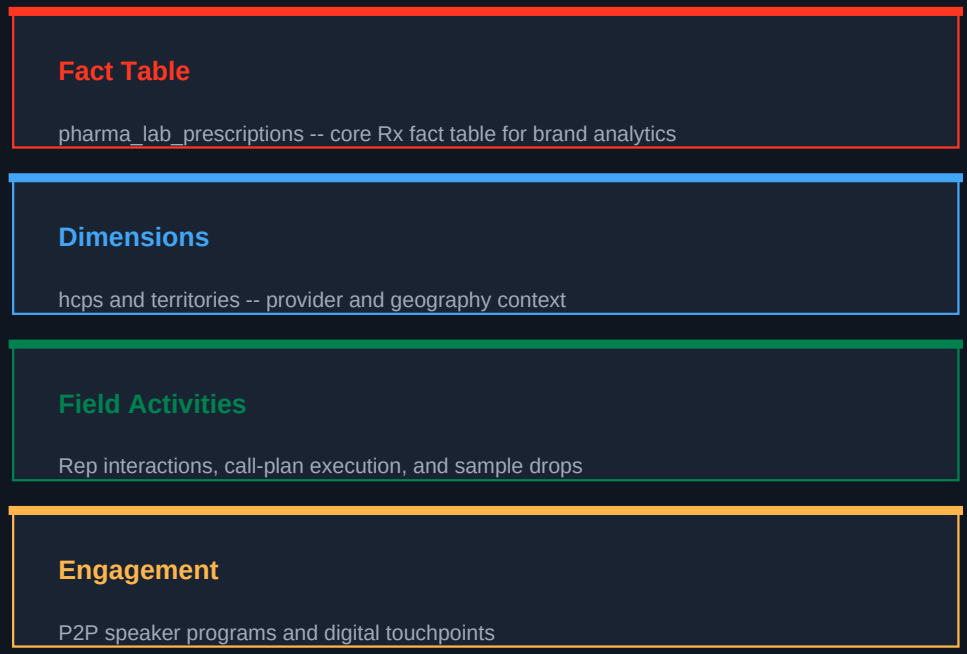
MODULE 4

Understanding the Source Data

Before building any Genie Agent (Space), understand the data the same way a commercial analytics user will.

Questions to Answer Up Front

- Which table answers prescriber volume questions?
- Which table answers field force productivity questions?
- Which columns hold business-friendly labels vs. technical IDs?
- Which metrics require joins or business rules?
- Which columns need synonyms?
- Which terms are ambiguous and need instruction?



MODULE 5

Genie Agent (Space) Design Approach

Build deliberately. Reliability comes from iteration, not one-time setup.

Choose Business Scope

Brand performance, field force productivity, omnichannel effectiveness, or executive portfolio analytics.

Identify Trusted Sources

Avoid mixing exploratory or duplicate tables without clear ownership.

Define Benchmarks First

Establish the questions the Agent (Space) must answer correctly before tuning.

Document Business Logic

Rx funnel ratios, NBRx definitions, active base rules, and exclusions.

Curate Incrementally

Start with metadata, then add synonyms, examples, SQL logic, and instructions.

Test & Refine

Systematic testing and iterative refinement drive production quality.

MODULE 6

Creating the Genie Agent (Space)

Step-by-Step Walkthrough

1. Open AI/BI and create a new Genie Agent (Space)
2. Add the six lab tables from Unity Catalog
3. Set a clear business-oriented name and description
4. Verify all tables are browsable in the Agent (Space)
5. Ask two baseline prompts before any curation
6. Record baseline response quality gaps

Name:

Bayer US Pharma Commercial Analytics

Tables:

pharma_lab_hcps, pharma_lab_territories, pharma_lab_prescriptions, pharma_lab_field_activities, pharma_lab_speaker_programs, pharma_lab_digital_touchpoints

Baseline Prompts:

"How many paid Rx does Kerendia have?" | "Show call-plan execution rate by rep"

MODULE 7

Knowledge Store Curation Toolkit

Six curation layers that teach Genie to answer reliably. Failures happen when these layers are incomplete.

01 Metadata

Descriptions on tables and columns give Genie business meaning and context.

02 Synonyms

Map business phrases to data objects: "prescriber" <-> hcp_name.

03 Prompt Matching

Align common user phrasings with known business intent for consistent resolution.

04 SQL Logic

Curated SQL patterns for definitions that must always be computed consistently.

05 Example Queries

Show Genie what strong questions and correct answers look like.

06 Text Instructions

Clarify boundaries, exclusions, priorities, and response style.

Synonyms & Example Queries

Synonym Mapping (Lab 4)

Business Phrase	Maps To
Prescriber	hcp_name
Territory rep	rep_name
New-to-Brand / NBRx	rx_type = 'New-to-Brand'
Call plan completion	call_plan_status = 'Completed'
Specialty Rx	channel = 'Specialty Pharmacy'
Digital open rate	engagement_score
Speaker program	pharma_lab_speaker_programs

Rx Funnel Conversion

ROUND(100.0 * paid / total, 2) -- Submitted -> Paid -> Paid-Insured by brand

Call-Plan Execution Rate

ROUND(100.0 * completed / total, 2) -- by rep and territory

NBRx Volume by Brand

COUNT(*) WHERE rx_type = 'New-to-Brand' AND rx_status IN ('Paid','Paid-Insured')

Text Instructions

Add business-facing instructions inside the Genie Agent (Space) to clarify defaults, exclusions, and join priorities.

Default Rx Base

When a user asks for Rx volume, always filter to Paid and Paid-Insured status only. Exclude Submitted unless explicitly requested.

Starting Table

Use pharma_lab_prescriptions as the starting table for brand performance and Rx volume questions.

Join Priorities

Join to hcps for provider context and segmentation. Join to territories for geography and rep performance.

NBRx Definition

New-to-Brand (NBRx) means rx_type = 'New-to-Brand'. Do not include Switch or Refill in NBRx counts.

MODULE 8

Benchmarking Strategy

A production-ready Genie Agent (Space) should be validated against a benchmark suite before broad sharing.

Benchmark Categories

- Direct retrieval -- paid Rx counts, HCP counts, total Rx value
- Grouped analysis -- by brand, territory, channel, HCP segment
- Join reasoning -- Rx + HCP, activity + territory, HCP + digital
- Business logic -- Rx funnel conversion, NBRx, call-plan execution
- Ambiguous language -- prescriber, specialty, digital open rate, volume

Dimension	What Good Looks Like
Table selection	Genie starts from the appropriate fact table
Join path	No missing or duplicate joins
Metric correctness	Output matches the benchmark SQL
Prompt robustness	Similar phrasings yield the same answer
Governance	Output respects user entitlements

MODULE 9

Sharing & Governance

Genie does not bypass Unity Catalog. Governance is automatic, not optional.

Sharing a Genie Agent (Space)

1. Share with the intended user group
2. Publish through the business-facing access path
3. Make it discoverable in Databricks One
4. Provide a short usage guide with example prompts

- Table & column permissions -- enforced automatically
- Row & column masking -- policies applied where configured
- Metadata visibility -- rules inherited from UC
- Lineage & auditing -- full context preserved

MODULE 10

Monitoring & Continuous Improvement

A Genie Agent (Space) becomes reliable through iteration. Assign clear ownership and run a continuous feedback loop.

Ongoing Monitoring & Improvement Loop

1. Observe real user questions
2. Identify failed or weak responses
3. Add or refine metadata, synonyms, examples, SQL logic, or instructions
4. Re-run the benchmark suite
5. Share improvements back to users

- Common prompt patterns from commercial analytics users
- Unanswered or poorly answered questions
- Frequently used terms not yet captured as synonyms
- Logic drift when multiple phrasings should produce the same answer
- User feedback trends and recurring correction themes

SELF-ASSESSMENT

Production Readiness Checklist

Foundations (Modules 1-4)

- [ ] Explain what a Genie Agent (Space) is
- [ ] Understand the execution pattern
- [ ] Know which table answers which question
- [ ] Profiled data across Rx, field, digital

Design & Creation (Modules 5-6)

- [ ] Followed a structured design workflow
- [ ] Created Agent with all six tables
- [ ] Tested baseline prompts, documented gaps

Knowledge Store (Module 7)

- [ ] Metadata on all tables and key columns
- [ ] Synonyms for pharma business terms
- [ ] Canonical example queries
- [ ] Curated SQL logic for key metrics
- [ ] Text instructions for defaults

Quality & Ops (Modules 8-10)

- [ ] Ran benchmarks, compared expected vs actual
- [ ] Governance enforced via Unity Catalog
- [ ] Continuous improvement loop defined
- [ ] Ownership areas assigned

COURSE SUMMARY

Production-Ready Genie Agent Standard

A fully operational Genie Agent (Space) requires all eight components working together.

Trusted Governed Tables	**Clear Metadata**	**Business Synonyms**	**Curated Example Prompts**
Certified SQL Logic	**Text Instructions**	**Benchmark Suite**	**Feedback Loop**

END OF WORKSHOP

Thank You

You are now equipped to build, curate, benchmark, govern, and continuously improve a production-grade Databricks Genie Agent (Space) for Bayer US Pharma Commercial Analytics.

10 Modules | 9 Labs | 3 Brands | Production Ready

INSTRUCTOR

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